

## Case Study 2

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### USER

Will the caffeine with SMILES: "CN1C=NC2=C1C(=O)N(C)C(=O)N2C" and aspirin with smiles "CC(=O)Oc1ccccc1C(=O)O" pass the cell membrane?

### AGENT

Proposed roadmap to assess membrane permeation:

- 1) Compute key physicochemical properties (MW, LogP, TPSA, HBD/HBA) from the provided SMILES.
- 2) Run ADMET permeability predictions (Caco-2, HIA, BBB where available).
- 3) Integrate results to answer whether each compound is likely to pass cell membranes at physiological conditions.

Proceeding now with steps 1–2 using your SMILES while awaiting your confirmation.

### Tool calls:

→ calculate\_ligand\_info  
→ calculate\_ligand\_info  
→ admet\_predict  
→ admet\_predict

### TOOL RESULT

calculate\_ligand\_info

### TOOL RESULT

calculate\_ligand\_info

### TOOL RESULT

admet\_predict

### TOOL RESULT

admet\_predict

## AGENT

### Roadmap

- 1) Compute key physicochemical properties (MW, LogP, TPSA, HBD/HBA).
- 2) Run permeability-oriented ADMET predictions (HIA, Caco-2/PAMPA, BBB).
- 3) Integrate into a membrane-pass/likelihood call for each molecule.

Proceeding with steps 1–2 while awaiting your confirmation.

### Results and interpretation

- Caffeine (SMILES: CN1C=NC2=C1C(=O)N(C)C(=O)N2C)

- Properties: MW 194.2; cLogP -1.03; TPSA 61.8 Å<sup>2</sup>; HBD 0; HBA 6.

- Permeability predictions: High human intestinal absorption (~1.00); strong PAMPA permeability (high score); Caco-2 consistent with moderate–high passive diffusion; likely BBB-permeant.

- Call: Yes—caffeine readily crosses cell membranes by passive diffusion and can penetrate the CNS.

- Aspirin (SMILES: CC(=O)Oc1ccccc1C(=O)O)

- Properties: MW 180.2; cLogP 1.31; TPSA 63.6 Å<sup>2</sup>; HBD 1; HBA 3.

- Acid–base note: pKa ≈ 3.5; at physiological pH (~7.4) it is mostly deprotonated (anion), which reduces passive permeability.

- Permeability predictions: High human intestinal absorption (~1.00); PAMPA indicates limited passive permeability; Caco-2 consistent with moderate permeability; BBB penetration is expected to be limited in practice due to ionization.

- Call: Conditional—unionized aspirin can cross membranes, but at physiological pH it is largely ionized and crosses less efficiently than caffeine. It is well absorbed orally (helped by gastric absorption and some neutral fraction) but is generally a poorer membrane crosser and shows limited CNS penetration.

If you want, I can refine the assessment for a specific tissue and pH (e.g., cytosol pH 7.2, stomach pH 1–2) or consider transporter effects.